

2026 MIDDLE SCHOOL STEM CAMPS

PY-STEM

INSTRUCTOR GUIDE PACKET

Facilitation guides for every primary and backup activity.

IN THIS PACKET

PYS-01 Magnetic Capsule Maze Cup

PYS-02 Oobleck Armor Arena

PYS-03 Cardboard Automata Arcade

PYS-04 Stethoscope Sprint and Recovery

PYS-05 Reaction Time Combine

PYS-06 SONAR Slinky Showdown

PYS-07 Pinhole Precision Challenge

PYS-08 Low-Ropes Force Map Relay

PYS-09 Hovercraft Hockey Hackathon

PYS-10 Spectra Sleuth Showdown

PYS-11 BookBot Bin Logic Challenge

PYS-12 Accessibility Ramp Rescue Lab

PYB-01 Pencil Pressure Safety Lab

PYB-02 Domino Neuron Relay

PYB-03 Pulley Rescue Relay

PYB-04 Barcode Checksum Rescue

PYB-05 Code Break Cipher Relay

Magnetic Capsule Maze Cup

Steer a pill-camera without touching it

At a glance: Steer a magnetic pill-camera through a body maze without touching it. About 80 min; 4 to 6 teams.

MATERIALS

- 6 mm driver magnets in bottle caps; keep the tiny 2 by 1 mm magnets locked away
- maze boards
- steel washers, nuts, or paper clips
- timers
- barrier sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does moving the magnet slowly help?
- How does distance change the pull you feel?
- Where is remote actuation used in real medicine?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Fastest clean run	50
Low penalties	20
Magnetic-control explanation	20
Redesign	10
Total	100

SOURCES Science Museum Group Magnetic Maze

Oobleck Armor Arena

Armor from a liquid that hardens on impact

At a glance: Design armor with a material that acts like a liquid until impact. About 80 min; 4 to 6 teams.

MATERIALS

- cornstarch
- water
- zip bags
- cardboard sleeves, a target (coin stack or foam square), and a rigid lid for the press
- a sealed corn-allergy pad (teacher-made)
- spoons
- trays

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a fast press feel harder than a slow one?
- How little oobleck still protected the target?
- Where would shear-thickening armor be useful?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Protection result	40
Material efficiency	20
Explanation	20
Redesign	10
Cleanup	10
Total	100

SOURCES Science Buddies Oobleck; Exploratorium Ooze

Cardboard Automata Arcade

One crank, one clever motion

At a glance: Build a one-crank machine that performs a mini task. About 80 min; 4 to 6 teams.

MATERIALS

- cardboard
- skewers
- straws
- craft sticks
- tape
- staff-run low-temperature hot glue station
- scissors

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How did your cam shape change the motion?
- What caused jams, and how did you fix them?
- Where do cams show up in real machines?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Reliable motion	30
Cam or linkage complexity	20
Task success	20
Explanation	20
Aesthetics and build quality	10
Total	100

SOURCES Exploratorium Cardboard Automata

Stethoscope Sprint and Recovery

Build a listening tool, measure like a scientist

At a glance: Build a medical listening tool and use it like a sports scientist. About 80 min; 4 to 6 teams. Use radial wrist pulse only, never a neck pulse. Measurement and light activity are optional; follow health-plan restrictions and stop for dizziness, chest pain, unusual trouble breathing, or feeling unwell.

MATERIALS

- funnels
- flexible tubing
- tape
- balloons or membranes
- stopwatches
- sanitizing wipes

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- What change most improved your sound quality?
- Why does a good seal matter so much?
- Why must repeated pulse counts use the same method and conditions?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Sound quality	30
Measurement accuracy	25
Explanation	20
Redesign	15
Data clarity	10
Total	100

SOURCES Science Buddies Make a Stethoscope

Reaction Time Combine

Prove which strategy cuts your delay

At a glance: Train like a sports scientist and prove which strategy cuts delay. About 80 min; 4 to 6 teams.

MATERIALS

- meter sticks
- reaction-time strip (this packet)
- pencils
- score sheets
- clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why use the median instead of your fastest catch?
- Which strategy helped, and why might it?
- Where does reaction time matter in real life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Best median reaction time	35
Data quality	20
Strategy improvement	20
Explanation	15
Teamwork	10
Total	100

SOURCES Science Buddies Reaction Time

SONAR Slinky Showdown

Make invisible distance sensing visible

At a glance: Make invisible distance sensing visible with a slinky pulse. About 80 min; 4 to 6 teams. Goggles are required whenever a slinky is stretched; keep it flat, held at both ends, and no longer than about 3 m.

MATERIALS

- 4 to 6 metal slinkies; safety goggles: one pair per participant while any slinky is stretched
- floor tape
- station cards
- clipboards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Fixed protocol: predict each station first; keep the slinky flat, held, and under about 3 m; measure L and count full round trips N.
- Predict and test the fixed cards, then time N full round trips and calculate total distance = $2 \times L \times N$.
- Check units and arithmetic, then defend one station answer with an observation and the round-trip calculation.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How is a slinky pulse like a sound wave?
- Why does timing an echo give distance?
- Where are ultrasonic sensors used?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Model and predictions	35
Station performance	25
Explanation	20
Team communication	20
Total	100

SOURCES IIHS SONAR Slinky

Pinhole Precision Challenge

The sharpest image, no lens

At a glance: Build the sharpest camera image with no lens. About 80 min; 4 to 6 teams.

MATERIALS

- cardstock
- foil
- tape
- pushpins or paper clips
- wax paper (through-view screen)
- light source

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why is the image upside down?
- What did shrinking the hole do, and why?
- How is this related to your eye?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Image clarity	35
Concept explanation	25
Build quality	20
Redesign	10
Speed	10
Total	100

SOURCES Exploratorium Personal Pinhole Theater; NASA JPL Pinhole Camera

Low-Ropes Force Map Relay

Predict balance before the balance challenge proves you right

At a glance: Predict balance and stability before the balance challenge proves you right. About 80 min; 4 to 6 teams.

MATERIALS

- balance challenge cards (this packet)
- clipboards
- center-of-mass templates (this packet), a backless chair, painter tape, and a coin and small weight
- pencils

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Fixed protocol: predict before each of the four challenge cards; test next to a clear wall and spot the backless-chair stand.
- Rotate through the four fixed challenge cards and record the observed result plus one force map for each.
- Compare predictions with results and defend one center-of-mass explanation; do not invent a new pose or redesign round.
- Score, debrief with evidence, reset the station.

DEBRIEF

- When did your prediction match what you felt?
- How did lowering your center of mass help?
- Where else does balance physics matter?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct predictions	35
Force-map quality	25
Debrief explanation	20
Teamwork reflection	20
Total	100

SOURCES TeachEngineering center-of-mass and balance references

Hovercraft Hockey Hackathon

A puck that barely touches the floor

At a glance: Build a puck that moves by barely touching the floor. About 80 min; 4 to 6 teams. Only staff use low-temperature hot glue. Score glide as round(30 x team best / field best), capped at 30; score the five-shot target total as round(raw / 6), capped at 25.

MATERIALS

- recycled CDs or plastic discs
- pop-top bottle caps
- balloons
- staff-run low-temperature hot glue station
- targets
- floor tape

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does the air cushion reduce friction?
- Why is more lift not always better?
- Where are air bearings used in real life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Glide performance	30
Target competition score	25
Explanation	20
Redesign	15
Build quality	10
Total	100

SOURCES NASA JPL Hovering on a Cushion of Air; Science Buddies Hovercraft

Spectra Sleuth Showdown

Two white lights, two hidden color fingerprints

At a glance: Can two white lights hide different color fingerprints? About 80 min; 4 to 6 teams.

MATERIALS

- diffraction glasses or gratings
- white LED and neon or gas sources
- incandescent source if available
- spectrum cards
- museum clue sheets

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Fixed method: predict, observe, and sketch each mystery spectrum before consulting the reference cards.
- Observe, sketch, match, and justify each source using a smooth rainbow, broad band, or separate lines as evidence.
- Audit the matches against visible evidence, then connect a line spectrum to fireworks colors without flame tests.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How can two white lights have different spectra?
- What does a line spectrum tell you?
- How do chemists make firework colors?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct matches	40
Spectrum sketches	20
Explanation	20
Exhibit connection	20
Total	100

SOURCES Science History Institute Flash Bang Boom; Exploratorium Spectra

BookBot Bin Logic Challenge

Retrieve by address, not by conveyor race

At a glance: Retrieve the right book by using bin addresses, not a conveyor race. About 80 min; 4 to 6 teams. For this station, override the generic setup: one shared tabletop judging mat, DEPOT beside A1, one team at a time, orthogonal adjacent moves only, start and finish at DEPOT, and score legal moves before time. Normalize the efficiency score as $\max(0, 10 - (\text{team legal moves} - \text{card minimum}))$; use time only as a tie-breaker.

MATERIALS

- bin cards
- address labels
- order deck
- planning route mats: one per team; shared judging mat: one
- timers
- optional grabber tool

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Shared-mat rules: start and finish at DEPOT, move orthogonally, retrieve every listed address, and run one team at a time.
- Plan on team copies, then run the selected card on the shared judging mat while staff count legal moves.
- Compare legal moves with that card's verified optimum, compute normalized efficiency, and defend the routing rule.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why store by address instead of by subject?
- What made one route use fewer legal moves?
- Where else is this storage logic used?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct retrievals	35
Legal complete route	25
Routing or algorithm explanation	20
Efficiency versus card optimum	10
Teamwork	10
Total	100

SOURCES Temple Charles Library BookBot; Charles Library Circulating Collections

Accessibility Ramp Rescue Lab

A portable ramp that meets real constraints

At a glance: Design a portable ramp that works for a real user constraint. About 80 min; 4 to 6 teams.

MATERIALS

- foam board
- craft sticks
- dowels
- binder clips
- rulers
- free-rolling or pull-back car and a 200 g hanging weight set
- tape

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does a gentler slope need more length?
- Which criterion was hardest to meet?
- Why design for accessibility from the start?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Load held	30
Slope meets target	30
Portability	20
Client criteria explanation	20
Total	100

SOURCES TeachEngineering Wheelchair Ramp Design

BACKUP ACTIVITIES

Ready alternates

Pencil Pressure Safety Lab

Spread a force before it dents the target

At a glance: Spread a force before it dents the target. About 60 min; 4 to 6 teams.

MATERIALS

- Pencils
- Foam
- Cardboard
- Weights
- Graph paper

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why does the same force do less damage over more area?
- Where is this used in safety gear?
- What made your spreader better?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Pressure spread	40
Explanation	25
Data quality	20
Redesign	15
Total	100

SOURCES IIHS Pencil Pressure

Domino Neuron Relay

Test a domino relay across a gap and name the model's limits

At a glance: Test a domino relay across a gap and name the model's limits. About 60 min; 4 to 6 teams.

MATERIALS

- Dominoes: about 16 tiles for one shared backup track
- Rulers: shared at the backup track
- Tape
- Index cards

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- What happened when the gap got too wide?
- Why is the bridge only an engineering repair rather than a literal synapse or myelin sheath?
- Why does myelin speed real signals?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Transmission success	35
Gap challenge	25
Explanation	20
Redesign	20
Total	100

SOURCES Exploratorium Domino Effect

Pulley Rescue Relay

Lift the load with smarter force, not stronger arms

At a glance: Lift the load with smarter force, not stronger arms. About 60 min; 4 to 6 teams.

MATERIALS

- Classroom pulleys
- String
- Weights
- Spring scales

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- Constraints: approved materials, one variable at a time, record before redesign.
- Build or solve, then run the standard test for a baseline.
- Redesign one variable, re-test, compare to baseline.
- Score, debrief with evidence, reset the station.

DEBRIEF

- Why did the movable pulley reduce effort?
- What did you trade for the easier lift?
- Where are pulley systems used in rescue work?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Performance	35
Setup accuracy	25
Explanation	20
Teamwork	20
Total	100

SOURCES TryEngineering Pulleys and Force

Barcode Checksum Rescue

Catch the fake barcode before the wrong part ships

At a glance: Catch the fake barcode before the warehouse ships the wrong part. About 60 min; 4 to 6 teams.

MATERIALS

- Printed barcode cards
- Dry erase boards
- Timers

FACILITATION ARC

- Hook and prediction: pose the mission, teams write one prediction and reason.
- UPC-A rule: verify 12 digits from the left, multiply odd positions by 3, and accept only totals that are multiples of 10.
- Verify and classify 12-digit UPC-A codes, recording correct detections and false alarms from the staff key.
- Audit flagged cards, show one complete weighted sum, and explain why a single changed digit is detected.
- Score, debrief with evidence, reset the station.

DEBRIEF

- How does one extra digit catch a typo?
- Why do false alarms matter in a warehouse?
- Where do you see check digits in daily life?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
Correct detections	45
Low false alarms	20
Rule explanation	20
Teamwork	15
Total	100

SOURCES Computer science unplugged error-detection concepts

Code Break Cipher Relay

Crack ten ciphers and race the answer to the judge

At a glance: Crack ten ciphers and race the answer to the judge before time runs out. Plan about 60 minutes for the core block; 4 to 6 teams of 4 to 5 students.

MATERIALS

- Printed decoder sheets (six, one per team)
- Printed code slips (ten pages, each cut into six slips)
- Staff-only answer key (in the PY-STEM instructor answer key pack; never posted)
- Pencils and scratch paper
- Timer

FACILITATION ARC

- Hook and decoder tour: pose the mission, walk the four tables and four letter rules.
- Rounds 1 to 5: one slip per team face down, GO, four minutes; log finish order.
- Runner rule: whisper or show it written; "not yet" on a miss, unlimited retries.
- Rounds 6 to 10: harder codes; the finale hides two codes on one slip.
- Score, debrief with evidence, collect slips and decoder sheets.

DEBRIEF

- Which code fell fastest once you knew the rule, and why?
- How did splitting the symbols across teammates change your speed?
- Where does the real world still use codes like these?

SCORING (100 POINTS)

SCORING CRITERION	POINTS
First correct team on a code	10
Second to fifth correct team	8, 6, 4, 2
Any later correct answer before time	1
Unsolved when time is called	0
Total · perfect sweep of ten codes	100

SOURCES Classic classroom ciphers (Caesar, Atbash, Polybius, Morse); computer science unplugged decoding relays